Amazon Web Services: Automation

Topics

- Why Do We Automate?
- CloudFormation
- Elastic Beanstalk
- Systems Manager

Why Do We Automate?

Why Use Automation?

- Manual builds are a gamble
- The best-case scenario is that you'll build it correctly, but you're more likely to make some costly errors
- This is where being lazy is a good thing
- Allows you to create and destroy your architecture as needed

Benefits of Automation

Time
We have more important things to do at work than manual processes

Consistency
When you automate, you get the same results every single time

Security
Easier to prevent security incidents, and if one occurs you can quickly fix it

AWS Automation Services

AWS CloudFormation

Allows you to provision resources quickly and consistently, and manage them throughout their lifecycles

AWS Elastic Beanstalk

Easy-to-use all in one service for deploying and scaling web applications and services developed with a variety of supported languages

AWS Systems Manager

Gives you the ability to easily patch, update, manage, and configure your EC2 instances along with on-premise architecture

AWS CloudFormation

What is AWS CloudFormation?

- An AWS service that allows us to declare our AWS infrastructure as code (IAC)
- Define via a JSON or YAML template
- Most AWS resources are supported but some are not

Process Overview

```yaml
AWSTemplateFormatVersion: 2010-09-09
Resources:
  VPC:
    Type: 'AWS::EC2::InternetGateway'
    Properties:
      Tags:
          - Key: Application
            Value: !Ref 'AWS::StackName'
          - Key: Network
            Value: Public

      GatewayToInternet:
        Type: 'AWS::EC2::VPCGatewayAttachment'
        Properties:
          VpcId: !Ref VPC
          InternetGatewayId: !Ref InternetGateway
```

Step 1

- Write you code
- Declarative programming language that supports either JSON or YAML formatting

Step 2

- Deploy your code as a Stack or Stack Set
- CloudFormation will make the needed AWS API calls on your behalf

CloudFormation Benefits

Infrastructure as Code
Allows you to use version control for all supported AWS infrastructure resources

Cost Tracking
Use stack tags to identify resources and their cost

Automated Deployment
No more manually standing up resources. Deploy and destroy them automatically via the template

Stacks Are Regional
Each deployed stack is a regional resources. Easily deploy portable stacks into other regions as needed

Change Sets
Can generate a preview of resource changes when updating an existing stack

Stack Sets
Use an admin account to create, update, or delete stacks across multiple accounts and regions in a single operation

Template Sections

AWSTemplateFormatVersion

Parameters

Mappings

Resources

Outputs

Transform

Optional

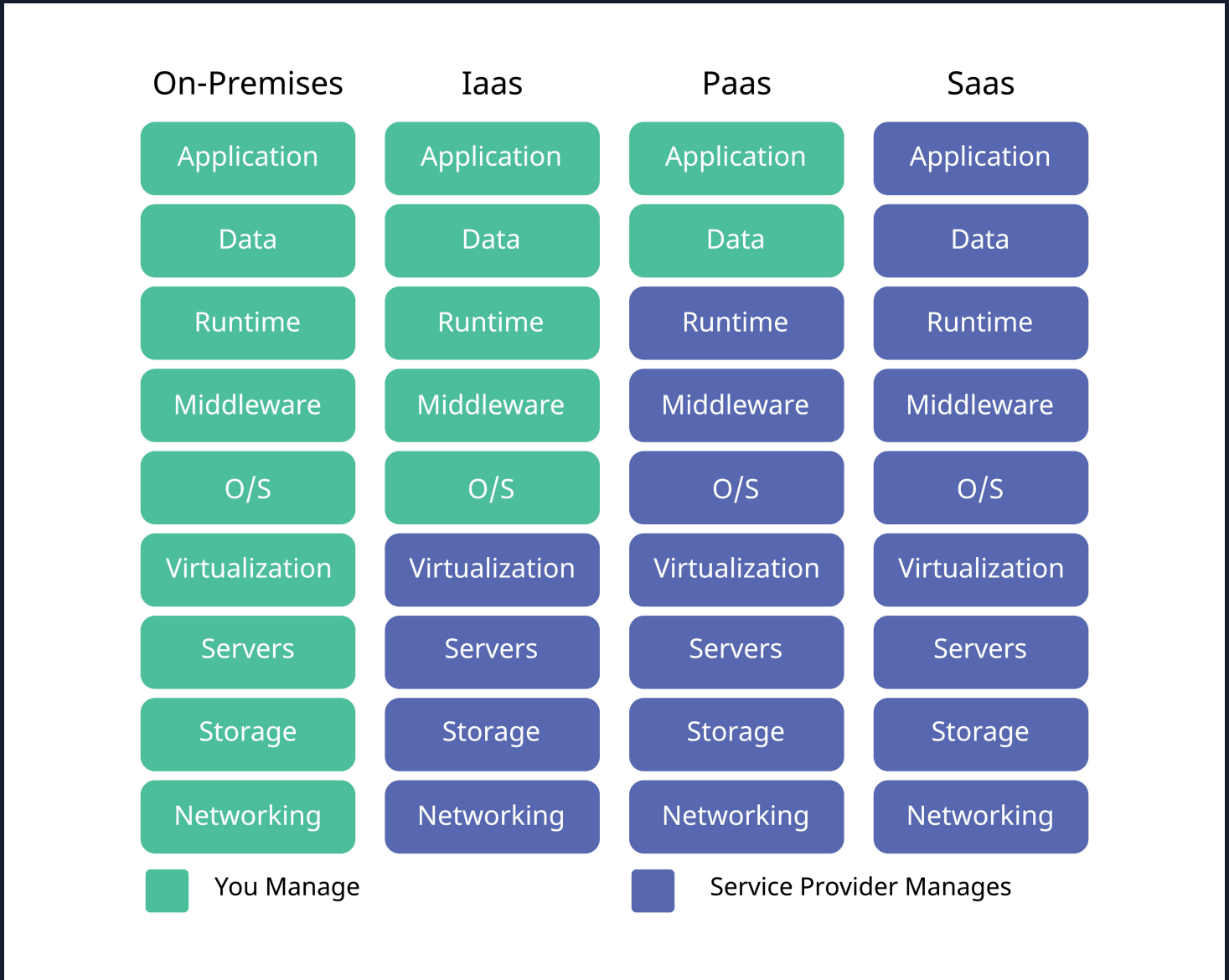
Required

Demo: Deploy CloudFormation Template

Elastic Beanstalk

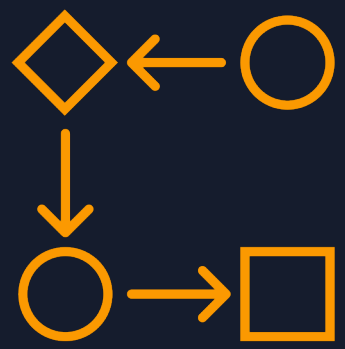
What is PaaS?

- Platform as a Service (PaaS) is a single-stop application deployment model
- You bring your code, and the provider builds everything for you, deploys your application, and then manages it going forward

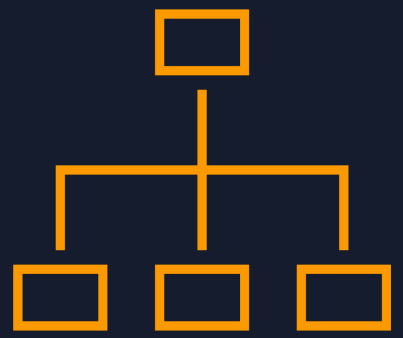


What is Elastic Beanstalk?

Automates all your deployments. Create a template of how you'd like your environment to look like



Handles all your deployments. You can upload your code, test your code, and then deploy to production



Handles building out the insides of your EC2 instances. You don't have to get on the hosts yourself

Systems Manager

What is System Manager?

- Systems Manager is a suite of tools designed to let you view, control, and automate both your AWS architecture and on-premises resources

- AWS Systems Manager Agent (SSM Agent) is a software component that is installed and then executed on managed instances within AWS as well as on premise

Important Capabilities

Capabilities

- **Automation:** Use predefined or custom playbooks (documents) to enable resource management
- **Run Command:** Remotely execute commands on managed compute without SSH or RDP
- **Patch Manager:** Automates patching managed instance (OS patches and applications)
- **Parameter Store:** Securely store your secrets and application config information
- **Maintenance Windows:** Define a schedule for performing actions on your managed instances
- **Sessions Manager:** Securely connect to your managed compute without needing SSH access

Session Manager Concepts

Logging

Allows you to log all usage during sessions (commands and connections) to CloudWatch and CloudTrail

SSM Agent

Supports both Linux and Windows without the need for SSH or RDP. Agent-based connection without opening ports

SSM Agent

Software that runs on compute (EC2, edge devices, on-prem servers, and custom VMs) that makes it possible for System Manager to update, manages and configure these resources

Typically preinstalled on most official AWS AMIs. Ensure that you have the IAM permissions needed, and it is running.

Possible to install the SSM agent on your own compute and edge devices to allow System Manager interactions

Parameter Store Concepts

Free feature to store config data and secret values in a hierarchical manner with parameter policies (expiration dates)

"ABC"

String: Parameters contain a block of plaintext

[A,B,C]

StringList: Parameters contain a comma-separated list of values

SecureString: Parameter where sensitive data needs to be stored and referenced in a secure manner (encrypted via AWS KMS)

Checkpoint

What are common use cases for the AWS Systems Manager Parameter Store "SecureString" parameter?

Choose 2

1. For sensitive data that needs to be stored and referenced in a secure manner.
2. For using data/parameters across AWS services without exposing the values as plaintext in commands, functions, agent logs, or CloudTrail logs.
3. For managing the patch level on your Windows servers.
4. For connecting to your EC2 instances.

Systems Manager can be used to patch and update _____ and _____.

1. Containers, ECS
2. Lambda, EC2
3. EC2 instances, on-premises instances
4. S3 buckets, EC2 instances

What part of a CloudFormation template allows you to pass values into the template?

1. Parameters
2. Conditions
3. Settings
4. Resources

Whenever possible, we should focus on _____ processes over doing things _____.

1. manually, automating
2. automating, manually
3. on premises installs, in the cloud
4. reviewing, at scale

Elastic Beanstalk is an example of _____.

1. Infrastructure as a Service (IaaS)
2. Unmanaged architecture
3. Amazon's terrible naming scheme
4. Platform as a Service (PaaS)

Your brand new CloudFormation stack just encountered an error. By default, what happens to the resources that it had created?

1. They are preserved
2. They are terminated.
3. They are reused when you try running the template again.
4. AWS leaves them running for 15 minutes to make sure you don't need anything from them.

Which of the following is NOT a benefit of deploying an application using Elastic Beanstalk?

1. Elastic Beanstalk will rewrite your applications from any language into Python.
2. You maintain full control over all of your architecture.
3. It will deploy and update your applications for you.
4. It will automatically replace instances that fail.

Amazon Web Services: Automation